

URL Shortener Monitoring Report

Generated: 1/27/2026, 3:57:44 AM
Powered by Cerebras AI (Llama 3.3 70B)

Executive Summary

Total URLs: 10288	Creation Rate: 0.00 URLs/sec
Redirect Rate: 0.00 req/sec	Error Rate: 0.0000 errors/sec
P50 Latency: 0.56ms	P95 Latency: 7.82ms
P99 Latency: 9.56ms	Click-Through Rate: 0.94%
Active Connections: 1	Database Size: 0.79 MB

Ø>Ý AI-Powered Analysis

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1. EXECUTIVE SUMMARY

The system health is rated as EXCELLENT due to 0% errors, good latency, and 100% uptime. The technical metrics indicate a stable and reliable system. Overall, the system is performing well, with no critical issues or major errors.

2. PERFORMANCE ANALYSIS

****Latency**:** The latency metrics show a median (P50) latency of 0.56ms, a 95th percentile (P95) latency of 7.82ms, and a 99th percentile (P99) latency of 9.56ms. These values indicate that the system is responding quickly to requests. The low latency values suggest that the system is well-optimized for performance.

****Throughput**:** The current usage levels are very low, with a URL creation rate and redirect rate of 0.000 URLs/sec and 0.000 requests/sec, respectively. This suggests that the system is not currently under heavy load. The system's resources are likely underutilized, given the low traffic volume.

****Reliability**:** The system's reliability is excellent, with a 100% success rate and zero errors. This indicates that the system is functioning correctly and is able to handle requests without issues. The lack of errors suggests that the system is stable and well-tested.

3. ACTIONABLE RECOMMENDATIONS

- [PRIORITY: Medium] Implement additional monitoring tools to collect more detailed metrics on system performance and latency.
- [PRIORITY: Low] Conduct a code review to identify potential areas for optimization and improvement.
- [PRIORITY: High] Develop a scaling plan to prepare for potential increases in traffic volume.
- [PRIORITY: Medium] Implement automated testing to ensure the system remains reliable and stable.
- [PRIORITY: Low] Explore opportunities to reduce database size and improve storage efficiency.
- [PRIORITY: Medium] Develop a backup and disaster recovery plan to ensure business continuity.

4. CAPACITY PLANNING

The system currently has a significant amount of headroom for growth, with very low usage levels and minimal resource utilization. The database size is relatively small, at 0.79 MB, and the active connections are minimal, at 1. This suggests that the system can handle a substantial increase in traffic volume before requiring additional resources.

As the system grows, it will be essential to monitor key metrics, such as latency, error rates, and resource utilization, to determine when to scale. Specific thresholds, such as a 20% increase in latency or a 10% increase in error rates, can be used to trigger scaling actions. Additionally, the system should be designed to scale horizontally, allowing for the easy addition of new resources as needed.

5. RISK ASSESSMENT

- Insufficient backups and disaster recovery planning pose a risk to business continuity.
- Lack of redundancy in critical system components may lead to single points of failure.
- Inadequate security measures may expose the system to potential threats and vulnerabilities.

- Incomplete monitoring and logging may make it difficult to detect and respond to issues.
- Dependence on a single database may lead to data loss or corruption in the event of a failure.

6. STRENGTHS & OPPORTUNITIES

- The system's low latency and high success rate indicate a well-optimized and reliable platform.
- Opportunities exist to improve monitoring and logging capabilities, allowing for more detailed insights into system performance.
- The system's current underutilization presents an opportunity to optimize resource allocation and improve efficiency, reducing costs and environmental impact.

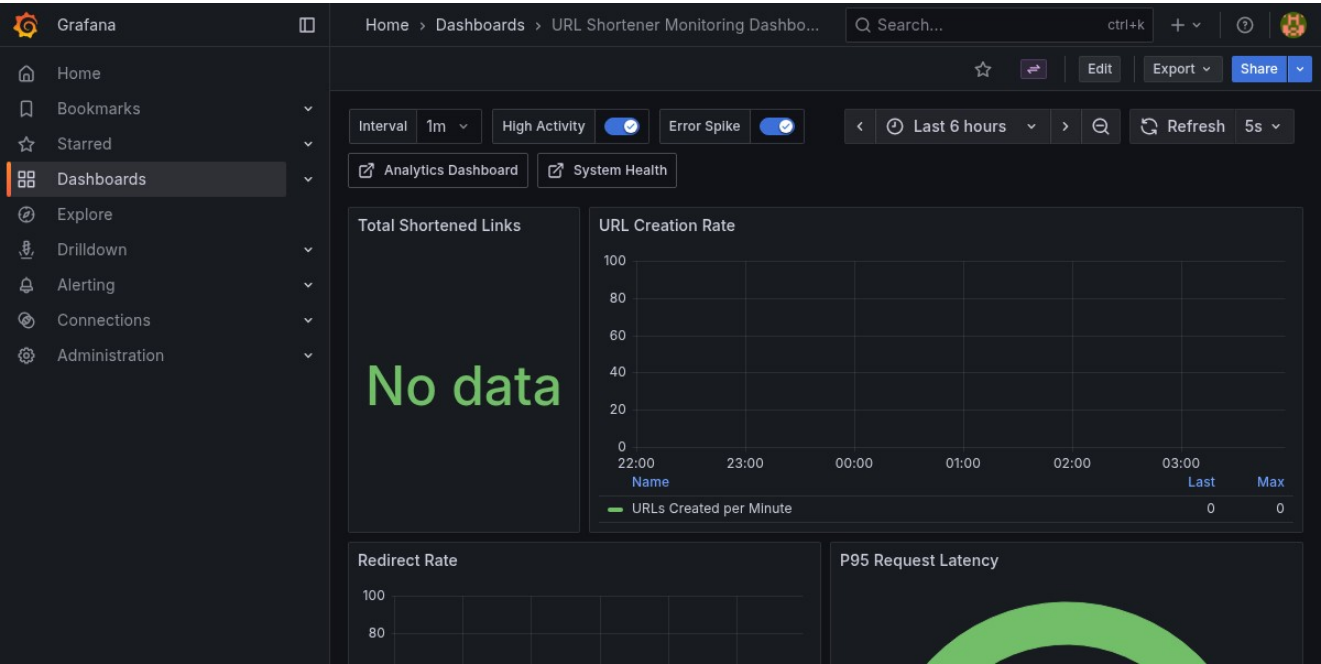
Ø=ÜÊ Detailed Metrics

Cumulative Statistics

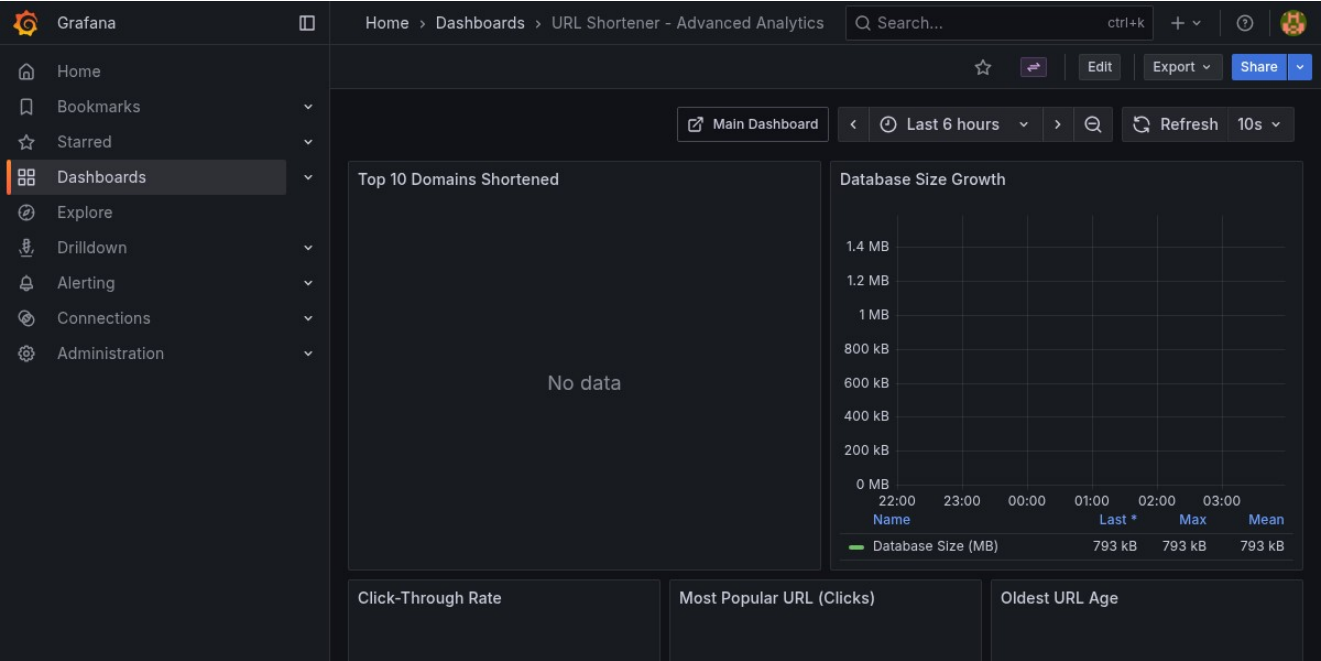
Total URLs Shortened: 0
Total Redirects: 0
Total 404 Errors: 0
Success Rate: NaN%

Ø=ÜÈ Dashboard Snapshots

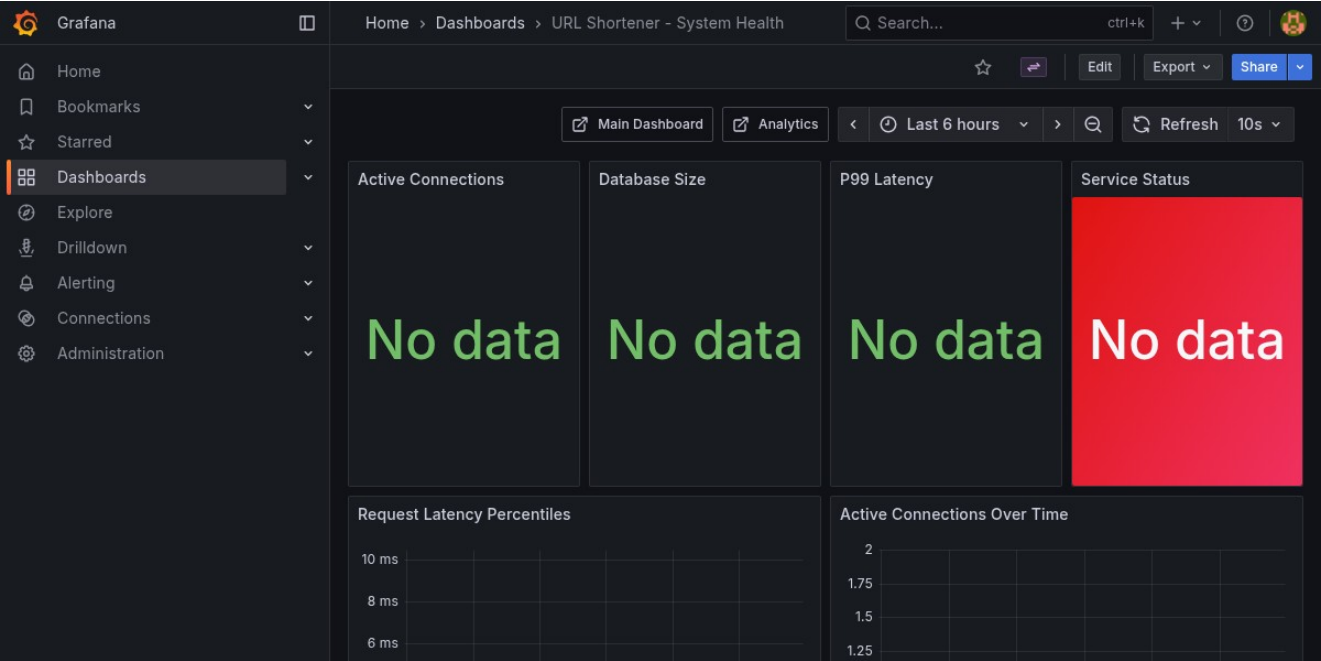
Main Monitoring Dashboard



Advanced Analytics Dashboard



System Health Dashboard



Live dashboards are available at:

- Main Monitoring: <http://localhost:3001/d/url-shortener-main>
- Advanced Analytics: <http://localhost:3001/d/url-shortener-analytics>
- System Health: <http://localhost:3001/d/url-shortener-system-health>

